In [3]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("ecommerce_sales_raw.csv")

df["Order_Date"] = pd.to_datetime(
    df["Order_Date"], dayfirst=True
)

print(df.head())
```

```
  Order_ID Order_Date Product_Name  Quantity       City Payment_Method  \
0  ORD0001 2025-04-13       Laptop         1    Chennai    Net Banking
1  ORD0002 2026-03-12   Headphones         3     Mumbai            UPI
2  ORD0003 2025-09-28      Speaker         1      Delhi            COD
3  ORD0004 2025-04-17       Tablet         5     Mumbai    Credit Card
4  ORD0005 2025-03-13      Monitor         2  Bengaluru    Credit Card

  Order_Status  Discount_Percent  Unit_Price     Category
0    Delivered                 0       65000  Electronics
1     Returned                 5        2500  Accessories
2     Returned                 5        4000  Electronics
3    Delivered                 5       22000  Electronics
4    Delivered                10       12000  Electronics
```

In [4]:
```
print(df.columns)
```

```
Index(['Order_ID', 'Order_Date', 'Product_Name', 'Quantity', 'City',
       'Payment_Method', 'Order_Status', 'Discount_Percent', 'Unit_Price',
       'Category'],
      dtype='str')
```

In [5]:
```
print(df["Product_Name"].value_counts())
```

```
Product_Name
Laptop        3
Speaker       2
Tablet        2
Keyboard      2
Headphones    1
Monitor       1
Webcam        1
Smartwatch    1
Smartphone    1
Name: count, dtype: int64
```

In [6]:
```
revenue_by_region = df.groupby("City")["Revenue"].sum()

print(revenue_by_region)
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[6], line 1
----> 1 revenue_by_region = df.groupby("City")["Revenue"].sum()
      2
      3 print(revenue_by_region)

File ~\Desktop\eda\.venv\Lib\site-
packages\pandas\core\groupby\generic.py:3002, in
DataFrameGroupBy.__getitem__(self, key)
   2995 if isinstance(key, tuple) and len(key) > 1:
   2996     # if len == 1, then it becomes a SeriesGroupBy and this is
actually
   2997     # valid syntax, so don't raise
   2998     raise ValueError(
   2999         "Cannot subset columns with a tuple with more than one
element. "
   3000         "Use a list instead."
   3001     )
→ 3002 return super().__getitem__(key)

File ~\Desktop\eda\.venv\Lib\site-packages\pandas\core\base.py:241, in
SelectionMixin.__getitem__(self, key)
    239 else:
    240     if key not in self.obj:
⟶ 241         raise KeyError(f"Column not found: {key}")
    242     ndim = self.obj[key].ndim
    243     return self._gotitem(key, ndim=ndim)

KeyError: 'Column not found: Revenue'
```

In [7]:
```
df["Revenue"] = (
    df["Quantity"] *
    df["Unit_Price"] *
    (1 - df["Discount_Percent"] / 100)
)

print(df[["Product_Name", "Quantity", "Unit_Price",
          "Discount_Percent", "Revenue"]])
```

	Product_Name	Quantity	Unit_Price	Discount_Percent	Revenue
0	Laptop	1	65000	0	65000.0
1	Headphones	3	2500	5	7125.0
2	Speaker	1	4000	5	3800.0
3	Tablet	5	22000	5	104500.0
4	Monitor	2	12000	10	21600.0
5	Tablet	2	22000	20	35200.0
6	Webcam	2	3500	20	5600.0
7	Keyboard	3	1500	0	4500.0
8	Laptop	5	65000	0	325000.0
9	Laptop	1	65000	5	61750.0
10	Speaker	4	4000	0	16000.0
11	Keyboard	1	1500	10	1350.0
12	Smartwatch	4	7000	20	22400.0
13	Smartphone	1	30000	5	28500.0

In [8]:
```
revenue_by_region = df.groupby("City")["Revenue"].sum()

print(revenue_by_region)
```

```
City
 chennai          35200.0
Bengaluru         21600.0
Chennai           65000.0
Delhi              3800.0
Hyderabad          4500.0
Mumbai           111625.0
Pune             325000.0
Vijayawada        39750.0
Visakhapatnam     95850.0
Name: Revenue, dtype: float64
```

In [9]:
```
region_summary = df.groupby("City").agg(
    Average_Revenue=("Revenue", "mean"),
    Order_Count=("Order_ID", "count"),
    Total_Revenue=("Revenue", "sum")
)

region_summary = region_summary.sort_values(
    by="Total_Revenue",
    ascending=False
)

print(region_summary)
```

```
              Average_Revenue  Order_Count  Total_Revenue
City
Pune                 325000.0            1       325000.0
Mumbai                55812.5            2       111625.0
Visakhapatnam         31950.0            3        95850.0
Chennai               65000.0            1        65000.0
Vijayawada            13250.0            3        39750.0
 chennai              35200.0            1        35200.0
Bengaluru             21600.0            1        21600.0
Hyderabad              4500.0            1         4500.0
Delhi                  3800.0            1         3800.0
```

In [10]:
```
units_sold = df.groupby("Product_Name")["Quantity"].sum()

print(units_sold.sort_values(ascending=False))
```

```
Product_Name
Laptop        7
Tablet        7
Speaker       5
Smartwatch    4
Keyboard      4
Headphones    3
Monitor       2
Webcam        2
Smartphone    1
Name: Quantity, dtype: int64
```

In [11]:
```
numeric_columns = df.select_dtypes(
    include=np.number
).columns

c_table = df[numeric_columns].corr()

print(c_table)
```

```
                  Quantity  Discount_Percent  Unit_Price   Revenue
Quantity          1.000000         -0.118644   -0.063237  0.523170
Discount_Percent -0.118644          1.000000   -0.361061 -0.331975
Unit_Price       -0.063237         -0.361061    1.000000  0.674244
Revenue           0.523170         -0.331975    0.674244  1.000000
```

In [12]:
```
correlation = df["Quantity"].corr(df["Revenue"])

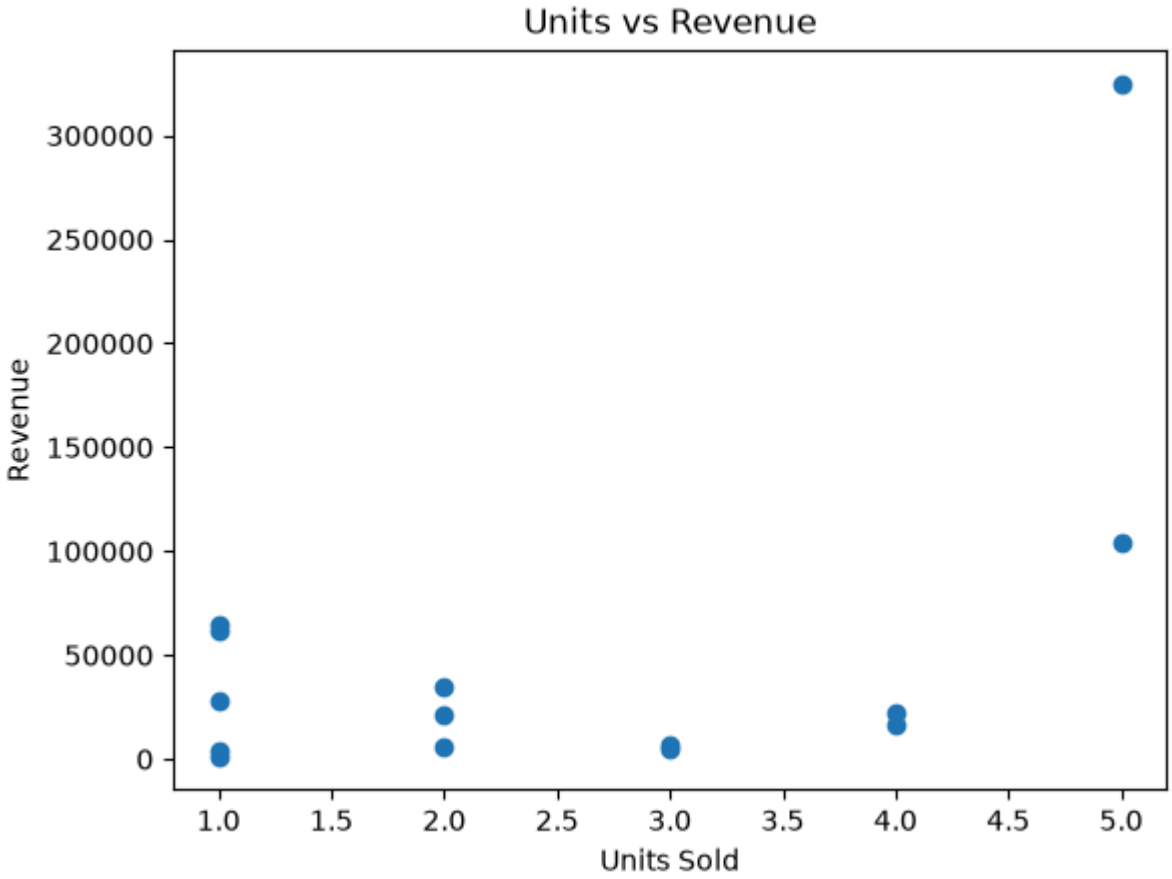
print("Units and Revenue Correlation:", correlation)
```

```
Units and Revenue Correlation: 0.5231696871364978
```

In [13]:
```
plt.scatter(df["Quantity"], df["Revenue"])

plt.xlabel("Units Sold")
plt.ylabel("Revenue")
plt.title("Units vs Revenue")

plt.show()
```



In [14]:
```
df["Month"] = df["Order_Date"].dt.month

monthly_revenue = df.groupby("Month")["Revenue"].sum()

print(monthly_revenue)
```

```
Month
1      63700.0
3      51125.0
4     495850.0
5       4500.0
8      61750.0
9       9400.0
11     16000.0
Name: Revenue, dtype: float64
```

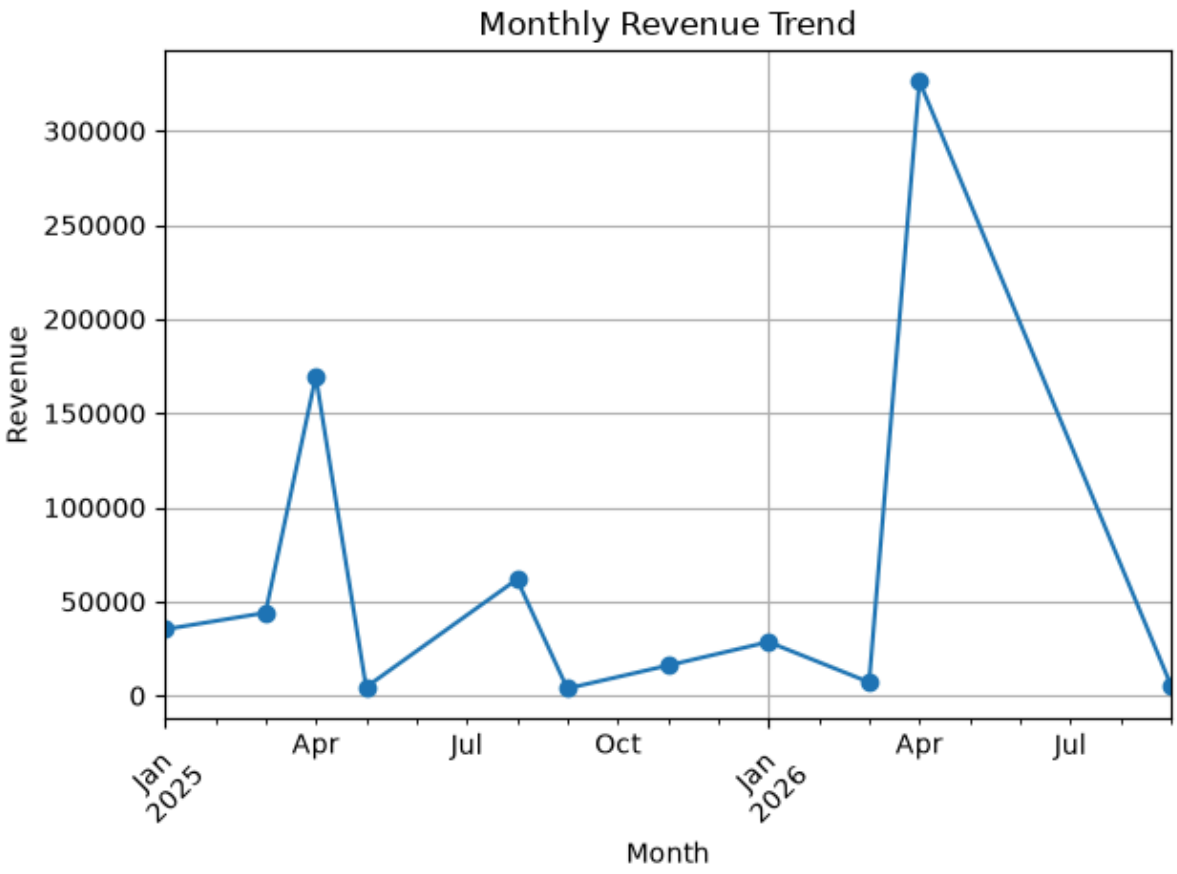
In [15]:
```
df["Month"] = df["Order_Date"].dt.to_period("M")

monthly_revenue = df.groupby("Month")["Revenue"].sum()

monthly_revenue.plot(kind="line", marker="o")

plt.xlabel("Month")
plt.ylabel("Revenue")
plt.title("Monthly Revenue Trend")
plt.xticks(rotation=45)
plt.grid(True)

plt.tight_layout()
plt.show()
```

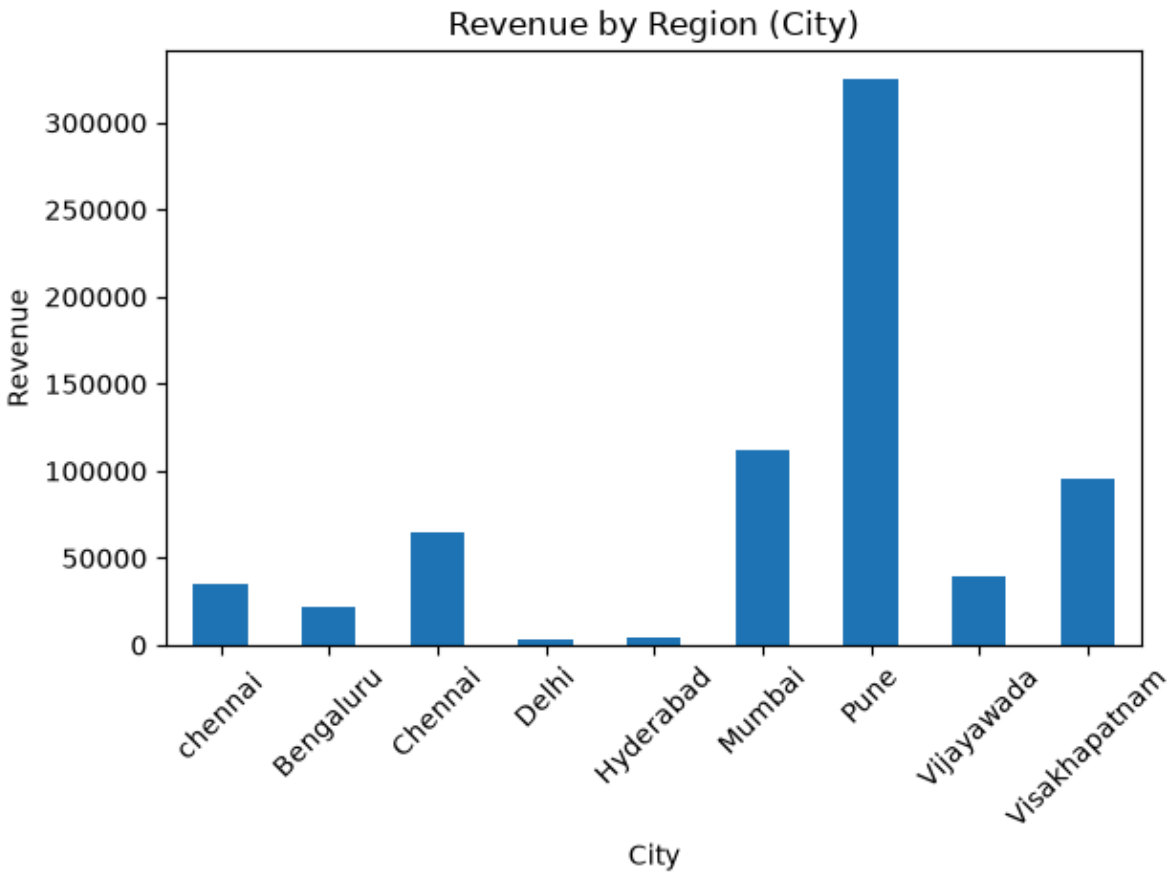


In [16]:
```
region_revenue = df.groupby("City")["Revenue"].sum()

region_revenue.plot(kind="bar")

plt.xlabel("City")
plt.ylabel("Revenue")
plt.title("Revenue by Region (City)")
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```



In [18]:
```
highest_revenue = df.sort_values(
    by="Revenue",
    ascending=False
)

print(highest_revenue[
    ["Order_ID", "Product_Name", "City", "Revenue"]
]).head(5)
```

	Order_ID	Product_Name	City	Revenue
8	ORD0009	Laptop	Pune	325000.0
3	ORD0004	Tablet	Mumbai	104500.0
0	ORD0001	Laptop	Chennai	65000.0
9	ORD0010	Laptop	Visakhapatnam	61750.0
5	ORD0006	Tablet	chennai	35200.0
13	ORD0014	Smartphone	Visakhapatnam	28500.0
12	ORD0013	Smartwatch	Vijayawada	22400.0
4	ORD0005	Monitor	Bengaluru	21600.0
10	ORD0011	Speaker	Vijayawada	16000.0
1	ORD0002	Headphones	Mumbai	7125.0
6	ORD0007	Webcam	Visakhapatnam	5600.0
7	ORD0008	Keyboard	Hyderabad	4500.0
2	ORD0003	Speaker	Delhi	3800.0
11	ORD0012	Keyboard	Vijayawada	1350.0

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[18], line 8
      4 )
      5
      6 print(highest_revenue[
      7     ["Order_ID", "Product_Name", "City", "Revenue"]
----> 8 ]).head(5)

AttributeError: 'NoneType' object has no attribute 'head'
```

In [19]:
```
highest_revenue = df.sort_values(
    by="Revenue",
    ascending=False
)

print(highest_revenue[
    ["Order_ID", "Product_Name", "City", "Revenue"]
])
```

```
   Order_ID Product_Name           City   Revenue
8   ORD0009       Laptop           Pune  325000.0
3   ORD0004       Tablet         Mumbai  104500.0
0   ORD0001       Laptop        Chennai   65000.0
9   ORD0010       Laptop  Visakhapatnam   61750.0
5   ORD0006       Tablet       chennai    35200.0
13  ORD0014   Smartphone  Visakhapatnam   28500.0
12  ORD0013   Smartwatch     Vijayawada   22400.0
4   ORD0005      Monitor      Bengaluru   21600.0
10  ORD0011      Speaker     Vijayawada   16000.0
1   ORD0002   Headphones         Mumbai    7125.0
6   ORD0007       Webcam  Visakhapatnam    5600.0
7   ORD0008     Keyboard      Hyderabad    4500.0
2   ORD0003      Speaker          Delhi    3800.0
11  ORD0012     Keyboard     Vijayawada    1350.0
```

In [20]:
```
print(highest_revenue.head(5))
```

```
  Order_ID Order_Date Product_Name  Quantity           City Payment_Method
\
8  ORD0009 2026-04-12       Laptop         5           Pune    Net Banking
3  ORD0004 2025-04-17       Tablet         5         Mumbai    Credit Card
0  ORD0001 2025-04-13       Laptop         1        Chennai    Net Banking
9  ORD0010 2025-08-03       Laptop         1  Visakhapatnam    Credit Card
5  ORD0006 2025-01-21       Tablet         2       chennai             COD

  Order_Status  Discount_Percent  Unit_Price     Category   Revenue    Month
8    Delivered                 0       65000  electronics  325000.0  2026-04
3    Delivered                 5       22000  Electronics  104500.0  2025-04
0    Delivered                 0       65000  Electronics   65000.0  2025-04
9      Shipped                 5       65000  Electronics   61750.0  2025-08
5    Delivered                20       22000  Electronics   35200.0  2025-01
```